

QUALITY EVALUATION OF PUPURU FORTIFIED WITH COCONUT BEETLE (*Oryctes rhinoceros*) LARVAE FLOUR

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Abstract

Pupuru, a fermented, smoke-dried cassava swallow food widely consumed in the riverine communities of southwestern and south-south Nigeria, is severely deficient in protein, with crude protein levels as low as 1.43 to 2.4%. This study evaluated the quality of pupuru fortified with flour from the larvae of the coconut rhinoceros beetle (*Oryctes rhinoceros*), an insect with well-documented nutritional richness that is culturally familiar to pupuru-consuming communities. Pupuru flour was blended with partially defatted, roasted coconut beetle larvae flour at six substitution levels: PF100 (control, 0%), PF95 (5%), PF90 (10%), PF85 (15%), PF80 (20%) and PF75 (25%). The blends were evaluated for proximate composition, mineral content, amino acid profile, anti-nutritional factors, pH and titratable acidity, functional properties, pasting properties, microbiological quality and sensory acceptability. Fortification raised crude protein more than fivefold, from 2.94% in the control to 16.34% in PF75, while increasing ash, fibre and fat and correspondingly reducing carbohydrate. Calcium, potassium, zinc and iron all increased significantly, with potassium rising nearly eightfold. The limiting amino acids of cassava, particularly lysine, methionine and tryptophan, were markedly enriched, raising the overall amino acid score from approximately 49% in the control to 73% at 25% substitution. Tannin and cyanogenic glycoside declined significantly with fortification, and cyanide levels in all blends remained far below the Codex safe limit. Functional and pasting properties shifted in a manner consistent with progressive starch dilution, while total bacterial rose and total fungal counts fell. Sensory acceptability declined significantly with increasing substitution, from 8.72 in the control to 4.82 in PF75. Integrating the nutritional, safety and sensory findings, the 10% substitution blend (PF90) offered the best balance between nutritional enhancement and consumer acceptability, more than doubling protein content while retaining an acceptability score corresponding to like slightly on the hedonic scale.

Keywords: Pupuru, coconut beetle flour, *Oryctes rhinoceros*, cassava fortification, edible insects, protein enrichment

Introduction

Food insecurity and protein-energy malnutrition remain among the most persistent public health challenges in sub-Saharan Africa, and Nigeria in particular. Approximately 33.1 million Nigerians are projected to face acute food insecurity during the 2025 lean season, a rise driven by economic hardship, record food inflation and ongoing insecurity in the northeast and northwest (World Food Programme, 2025), and the country was ranked 115th of 123 nations on the 2025 Global Hunger Index, reflecting a serious level of hunger. Cassava (*Manihot esculenta* Crantz) is a dietary cornerstone across the country, and among its many derived products, pupuru, a fermented, dewatered, moulded and smoke-dried cassava food, holds a distinctive place in the food culture of the riverine communities of Ondo, Delta, Edo and neighbouring Yoruba-, Ijaw- and Ilaje-speaking areas of southwestern and south-south Nigeria, where it is also known as ikwurikwu (Abiola *et al.*, 2021). Despite its cultural and dietary importance, pupuru remains, like most cassava products, a carbohydrate-dominant food with very poor protein content, typically ranging from 1.43% to 2.4% (Oladeji *et al.*, 2025; Abiola *et al.*, 2021), a level far below the recommended dietary allowance and a direct contributor to the stunting, wasting and severe acute malnutrition that continue to affect a large share of Nigerian children (UNICEF, 2024).

Previous efforts to close this protein gap through fortification with soy flour, African yam bean flour, moringa seed and sesame seed have improved the nutritional profile of pupuru to varying degrees, but often at the cost of elevated anti-nutrient levels, altered sensory character, or limited local affordability and availability (Oladeji *et al.*, 2025; Isaac-Bamgboye *et al.*, 2020; Jaho & Oluwamukomi, 2023). Edible insects present a promising alternative. The larvae of the coconut rhinoceros beetle, *Oryctes rhinoceros* (Linnaeus, 1758), commonly known as palm grubs, are a traditional delicacy in coastal and riverine communities where oil palm, coconut palm and raffia palm are abundant, the same communities that consume pupuru as a staple, giving the insect strong cultural and geographic relevance as a fortificant. *Oryctes rhinoceros* larvae have been reported to contain 34.76% to 44% crude protein on a dry matter basis, together with favourable amino acid and mineral profiles (Anaduaka *et al.*, 2021; Mwanza *et al.*, 2025; Egonyu *et al.*, 2024), and processing studies have shown that roasting improves both the protein content and the microbial safety of the larvae (Muthee *et al.*, 2024). Despite this well-documented nutritional richness and the well-established protein deficiency of pupuru, no published study has evaluated the quality of pupuru specifically fortified with coconut beetle larvae flour. This study therefore investigated the proximate, mineral, amino acid, anti-nutritional, physicochemical, functional, pasting, microbiological and sensory properties of pupuru fortified with coconut beetle flour at graded substitution levels, with the aim of identifying a formulation that meaningfully improves the nutritional quality of this staple food without compromising its safety or acceptability.

Materials and Methods

2.1 Materials

Fresh, mature cassava tubers of the TMS 419 variety were sourced from local farms in the riverine areas of Ondo State, Nigeria, where pupuru is traditionally produced and consumed. Third-instar larvae of *Oryctes rhinoceros* were collected from decomposing oil palm (*Elaeis guineensis*) and coconut palm (*Cocos nucifera*) stumps in plantations within the same locality, with species identity confirmed by an entomologist affiliated with the institution. All chemicals and reagents used for the proximate, physicochemical, mineral, amino acid, anti-nutritional and microbial analyses were of analytical grade.

2.2 Preparation of Pupuru Flour

Pupuru flour was produced by the traditional method described by Oladeji *et al.* (2025), with minor modification. Peeled, washed cassava tubers were submerged in water at a cassava-to-water ratio of 1:3 and allowed to ferment at ambient temperature (28 to 30 °C) for 72 hours, with the fermentation water changed once at 36 hours. The fermented cassava was dewatered under a wooden press for 6 to 8 hours, hand-moulded into cylindrical balls of 8 to 10 cm diameter, and smoke-dried over a wood fire for 6 to 8 hours, with periodic turning every 2 hours. The dried outer crust was scraped to remove ash and soot, and the inner dark-brown portion was milled and sieved through a 250 µm sieve to obtain pupuru flour, which was stored in sealed polyethylene bags at room temperature (25 ± 2 °C) until analysis.

2.3 Preparation of Coconut Beetle (*Oryctes rhinoceros*) Flour

Coconut beetle flour was prepared following the method of Muthee *et al.* (2024), with adaptation for flour production. Live larvae were held in moistened sawdust for 24 hours to allow gut evacuation, washed, blanched in boiling water (100 °C, 3 minutes) and roasted at 150 °C for 30 minutes, with turning every 10 minutes. After cooling, the wings, legs and chitinous head capsule were manually removed. The roasted larvae were partially defatted by n-hexane extraction in a Soxhlet apparatus for 4 hours to prevent excessive greasiness and rancidity while retaining nutritionally beneficial unsaturated fatty acids (Okaraonye & Ikewuchi, 2008; Nsevolo Miankeba *et al.*, 2022). The partially defatted material was milled and sieved through a 250 µm sieve and stored at 4 °C until use.

2.4 Formulation of Fortified Pupuru Blends

Pupuru flour (denoted P) and coconut beetle flour (denoted F) were combined by weight using a calibrated analytical balance (± 0.01 g) and homogenised by double sieving through a 250 µm sieve, following the approach of Adebayo *et al.* (2021). Six blends were formulated: PF100 (100% pupuru flour, control), PF95 (95:5), PF90 (90:10), PF85 (85:15), PF80 (80:20) and PF75 (75:25 pupuru to coconut beetle flour).

2.5 Proximate Composition

Moisture, crude protein (Kjeldahl, N × 6.25), crude fat (Soxhlet extraction), crude fibre and total ash were determined in duplicate using standard AOAC (2019) methods. Total carbohydrate was calculated by difference.

2.6 Physicochemical Properties

The pH of a 10% (w/v) aqueous dispersion of each blend was measured using a calibrated digital pH meter. Total titratable acidity was determined by titrating a filtered 10% aqueous extract against 0.1 N NaOH using phenolphthalein indicator, and expressed as percentage lactic acid (Oladeji *et al.*, 2025).

2.7 Functional Properties

Bulk density was determined by the tapped-cylinder method of Okaka and Potter (1979). Water solubility and swelling power were determined by the method of Leach *et al.* (1959) as adapted by Lerma *et al.* (2025). Water absorption capacity and oil absorption capacity were determined by the centrifugation method of Beuchat (1977), and least gelation temperature by the method of Coffman and Garcia (1977).

2.8 Pasting Properties

Pasting properties were determined using a Rapid Visco Analyser following ICC Method 162, with a programmed profile of heating from 50 °C to 95 °C, holding, and cooling back to 50 °C over a total run time of 23 minutes, as applied for cassava-based flours by Lerma *et al.* (2025) and Aidoo *et al.* (2022). Peak, trough, breakdown, final and setback viscosity, peak time and pasting temperature were recorded.

2.9 Anti-Nutritional Factors

Phytate was determined spectrophotometrically by the method of Wheeler and Ferrel (1971). Tannin was determined spectrophotometrically using tannic acid as standard (Trease & Evans, 1989). Cyanogenic glycoside was determined as hydrogen cyanide equivalent by alkaline distillation and titration against silver nitrate, following AOAC (2019) Method 915.03.

2.10 Mineral Composition

Calcium, potassium, zinc and iron were determined by atomic absorption spectrophotometry following dry ashing, according to AOAC (2019) Method 985.35.

2.11 Amino Acid Profile

Amino acid profiles of the control (PF100), the 10% blend (PF90) and the 25% blend (PF75) were determined by HPLC following acid hydrolysis (6 N HCl, 110 °C, 24 hours) and phenylisothiocyanate pre-column derivatisation, according to Spackman *et al.* (1958) as modified by Moore and Stein (1963). Tryptophan was determined separately following alkaline hydrolysis.

2.12 Microbial Analysis

Total bacterial count was determined by the pour plate method on Plate Count Agar (37 °C, 48 hours), and total fungal count by the spread plate method on acidified Potato Dextrose Agar (25 °C, 5 days), following AOAC (2019) and Muthee *et al.* (2024).

2.13 Sensory Evaluation

Sensory evaluation of the six cooked pupuru swallow samples was conducted by a semi-trained panel of 20 judges familiar with pupuru, using a nine-point hedonic scale (1 = dislike extremely, 9 = like extremely) to score appearance, aroma, taste, texture and overall acceptability.

2.14 Statistical Analysis

All data were subjected to one-way analysis of variance (ANOVA) using IBM SPSS Statistics version 26, and means were separated using Duncan's Multiple Range Test at $p \leq 0.05$. Proximate, physicochemical, functional, mineral, anti-nutritional and pasting results are reported as mean $\pm$ SD of duplicate determinations; sensory results are reported as mean $\pm$ SD of scores from 20 panelists.

3. Results and Discussion

3.1 Proximate Composition

Table 1 presents the proximate composition of the fortified pupuru blends. Moisture declined only marginally, from 5.33% in PF100 to 4.95% in PF80 ($P > 0.05$), a difference attributable to the lower hygroscopicity of the protein- and chitin-dominated insect flour relative to the hydroxyl-rich starch matrix of cassava; all values fell comfortably below the 10% threshold associated with microbiological stability in dry flours. Crude protein rose markedly and significantly, from 2.94% in PF100 to 16.34% in PF75 ($P < 0.05$), a 5.6-fold increase driven by the substitution of protein-poor cassava with the concentrated nitrogen source of *Oryctes rhinoceros* larvae, which has been reported to contain 42.29% crude protein (Egonyu et al., 2024). At the 10% inclusion level (PF90), protein reached 6.53%, closely corroborating the 6.46–6.81% reported by Ssepuuya *et al.* (2025) for cassava flour fortified with 8.36–10.52% house cricket powder. A 100 g portion of PF75 would supply approximately 29% of the adult daily protein requirement, against roughly 5% for the unfortified control (FAO/WHO/UNU, 2007).

Total ash rose significantly from 2.48% to 3.66% ($P < 0.05$), reflecting the substantial endogenous mineral load of insect tissue (Okaraonye & Ikewuchi, 2008; Mwanza *et al.*, 2025). Crude fibre increased from 1.71% to 2.92% ($P < 0.05$), attributable to chitin, the nitrogenous structural polysaccharide of the insect cuticle, which behaves as an insoluble dietary fibre and has documented prebiotic activity (Stull & Weir, 2023). Crude fat rose modestly from 0.84% to 2.04% ($P < 0.05$), a smaller increase than the 2.30–2.77% reported by Ssepuuya *et al.* (2025) for cricket-fortified cassava flour, consistent with the partial defatting applied to the larvae in this study. Carbohydrate declined correspondingly from 86.69% to 70.08%, an arithmetic consequence of proximate closure that is nutritionally favourable, as the shift from rapidly digestible starch toward protein and fibre would be expected to lower the glycaemic load of the reconstituted swallow (Abiola et al., 2021).

Table 1. Proximate composition of pupuru fortified with coconut beetle flour (% , mean $\pm$ SD, $n = 2$)

Sample	Moisture	Crude Protein	Total Ash	Crude Fibre	Crude Fat	Carbohydrate
PF100	5.33 ^a $\pm$ 0.02	2.94 ^c $\pm$ 0.04	2.48 ^c $\pm$ 0.04	1.71 ^c $\pm$ 0.02	0.84 ^{cd} $\pm$ 0.03	86.69 ^a $\pm$ 0.03
PF95	5.25 ^a $\pm$ 0.02	3.29 ^c $\pm$ 0.02	2.66 ^{bc} $\pm$ 0.01	1.94 ^{bc} $\pm$ 0.01	1.12 ^c $\pm$ 0.05	85.72 ^b $\pm$ 0.02
PF90	5.14 ^a $\pm$ 0.04	6.53 ^d $\pm$ 0.03	2.83 ^b $\pm$ 0.01	2.20 ^b $\pm$ 0.02	1.35 ^c $\pm$ 0.01	81.95 ^c $\pm$ 0.03
PF85	5.11 ^a $\pm$ 0.01	9.85 ^c $\pm$ 0.03	3.04 ^b $\pm$ 0.02	2.42 ^b $\pm$ 0.01	1.59 ^{ab} $\pm$ 0.01	78.00 ^d $\pm$ 0.06
PF80	4.95 ^a $\pm$ 0.03	13.13 ^b $\pm$ 0.05	3.54 ^a $\pm$ 0.02	2.66 ^a $\pm$ 0.01	1.83 ^a $\pm$ 0.01	73.89 ^e $\pm$ 0.06
PF75	4.97 ^a $\pm$ 0.06	16.34 ^a $\pm$ 0.01	3.66 ^a $\pm$ 0.02	2.92 ^a $\pm$ 0.02	2.04 ^a $\pm$ 0.02	70.08 ^f $\pm$ 0.07

Means within a column bearing different superscript letters differ significantly ($P < 0.05$). PF100 = 100% pupuru flour (control); PF95, PF90, PF85, PF80, PF75 = 95%, 90%, 85%, 80%, 75% pupuru flour with 5%, 10%, 15%, 20%, 25% coconut beetle flour respectively.

3.2 Mineral Composition

Table 2 shows that all four minerals increased substantially and significantly with fortification ($P < 0.05$). Calcium rose from 15.45 to 34.88 mg/100 g, and potassium showed the most dramatic enrichment of all measured parameters, rising nearly eightfold from 17.06 to 139.34 mg/100 g, consistent with the high intracellular potassium and calcium reported for *Oryctes* larvae (Mwanza *et al.*, 2025; Egonyu *et al.*, 2024). Zinc more than doubled, from 0.96 to 2.26 mg/100 g, reflecting zinc bound in insect metalloenzymes and cuticular protein. Iron rose from 3.76 mg/100 g in PF100 to a peak of 5.03 mg/100 g in PF80 before declining slightly to 4.64 mg/100 g in PF75, a reversal plausibly linked to iron chelation by the elevated chitin and residual phytate fraction at the highest inclusion level (Maiyo *et al.*, 2022). A 100 g portion of PF80 would supply approximately 72% of the daily iron requirement of a young child, against approximately 54% for the control, a nutritionally meaningful gain in a population with a high prevalence of iron-deficiency anaemia.

Table 2. Mineral composition of pupuru fortified with coconut beetle flour (mg/100 g, mean $\pm$ SD, $n = 2$)

Sample	Calcium (Ca)	Potassium (K)	Zinc (Zn)	Iron (Fe)
PF100	15.45 ^f $\pm$ 0.04	17.06 ^f $\pm$ 0.02	0.96 ^c $\pm$ 0.02	3.76 ^{bd} $\pm$ 0.01
PF95	19.53 ^e $\pm$ 0.03	38.92 ^e $\pm$ 0.05	1.37 ^b $\pm$ 0.01	3.97 ^{abc} $\pm$ 0.00
PF90	23.77 ^d $\pm$ 0.01	64.65 ^d $\pm$ 0.03	1.58 ^{ab} $\pm$ 0.02	4.15 ^b $\pm$ 0.04
PF85	28.63 ^c $\pm$ 0.01	91.07 ^c $\pm$ 0.03	1.91 ^a $\pm$ 0.03	4.54 ^{ab} $\pm$ 0.03
PF80	30.73 ^b $\pm$ 0.02	126.53 ^b $\pm$ 0.04	2.05 ^a $\pm$ 0.02	5.03 ^a $\pm$ 0.01
PF75	34.88 ^a $\pm$ 0.03	139.34 ^a $\pm$ 0.04	2.26 ^a $\pm$ 0.02	4.64 ^a $\pm$ 0.06

Means within a column bearing different superscript letters differ significantly ($P < 0.05$).

Sample keys as in Table 1.

3.3 Amino Acid Composition

Total amino acid content increased progressively from 63.50 g/100 g protein in PF100 to 73.69 g/100 g protein in PF90 and 84.00 g/100 g protein in PF75, with total essential amino acids rising correspondingly from 27.08 to 30.09 to 35.27 g/100 g protein (Table 3). Most nutritionally significant was the enrichment of the amino acids characteristically limiting in cassava-based staples: lysine rose from 2.84 to 4.14 g/100 g protein, methionine more than doubled from 0.72 to 1.63, tryptophan nearly tripled from 0.42 to 1.16, and threonine rose from 2.61 to 3.72 g/100 g protein, consistent with the favourable histidine, methionine and phenylalanine contribution previously reported for *Oryctes rhinoceros* larva (Okaraonye & Ikewuchi, 2008; Nsevolo Miankeba *et al.*, 2022). Scored against the FAO/WHO/UNU (2007) reference pattern for preschool children, the overall amino acid score of the protein improved from approximately 49% in the control, with tryptophan first-limiting, to approximately 73% in PF75, with lysine becoming first-limiting at a substantially higher score, indicating a marked improvement in overall protein quality with fortification.

Table 3. Amino acid composition of selected pupuru blends (g/100 g protein)

Amino acid	PF100 (control)	PF90 (10%)	PF75 (25%)
Leucine*	7.62	7.88	8.67
Lysine*	2.84	3.18	4.14
Isoleucine*	3.67	3.73	4.32
Phenylalanine*	3.19	3.64	3.99
Tryptophan*	0.42	0.84	1.16

Amino acid	PF100 (control)	PF90 (10%)	PF75 (25%)
Valine*	3.77	3.42	3.80
Methionine*	0.72	1.26	1.63
Histidine*	2.24	3.23	3.84
Threonine*	2.61	2.91	3.72
Arginine	3.87	4.13	4.30
Proline	3.66	4.16	4.57
Tyrosine	2.75	2.93	3.44
Cystine	1.15	1.45	2.06
Alanine	3.60	4.32	5.01
Glutamic acid	10.14	12.42	13.25
Glycine	2.92	3.61	3.97
Serine	3.00	3.35	4.00
Aspartic acid	5.33	7.23	8.13
Total essential amino acids	27.08	30.09	35.27
Total amino acids	63.50	73.69	84.00

Asterisks denote essential amino acids. Sample keys as in Table 1.

3.4 Anti-Nutritional Factors

All three anti-nutrients declined with fortification (Table 4), indicating that fortification improved the safety profile of the product. Phytate declined slightly and non-significantly from 0.72 to 0.54 mg/100 g ($P > 0.05$), a trend consistent with the phytate-reducing effect of the prolonged fermentation step in pupuru production (Jaho & Oluwamukomi, 2023). Tannin declined significantly from 54.23 to 42.76 mg/100 g ($P < 0.05$), reflecting dilution of the tannin-bearing cassava fraction, a reduction that is nutritionally beneficial given the protein-binding, digestibility-lowering action of tannins. Cyanogenic glycoside declined steeply and significantly, from 0.48 to 0.12 mg/kg HCN ($P < 0.05$); critically, every blend remained far below the 10 mg/kg safe limit set by the Codex Alimentarius Commission (2019), confirming that coconut beetle fortification enhances rather than compromises the cyanide-detoxifying effect of traditional pupuru fermentation.

Table 4. Anti-nutritional factors in pupuru fortified with coconut beetle flour (mean $\pm$ SD, $n = 2$)

Sample	Phytate (mg/100 g)	Tannin (mg/100 g)	Cyanogenic glycoside (mg/kg HCN)
PF100	0.72 ^a $\pm$ 0.01	54.23 ^a $\pm$ 0.04	0.48 ^a $\pm$ 0.01
PF95	0.67 ^a $\pm$ 0.03	50.73 ^b $\pm$ 0.08	0.42 ^a $\pm$ 0.02
PF90	0.62 ^a $\pm$ 0.04	47.88 ^c $\pm$ 0.06	0.34 ^b $\pm$ 0.01
PF85	0.61 ^a $\pm$ 0.01	45.55 ^c $\pm$ 0.23	0.23 ^c $\pm$ 0.05
PF80	0.58 ^a $\pm$ 0.04	44.19 ^{cd} $\pm$ 0.07	0.19 ^d $\pm$ 0.01
PF75	0.54 ^a $\pm$ 0.04	42.76 ^d $\pm$ 0.13	0.12 ^e $\pm$ 0.01

Means within a column bearing different superscript letters differ significantly ($P < 0.05$). Sample keys as in Table 1.

3.5 pH and Titratable Acidity

The pH and titratable acidity moved in the expected inverse relationship (Table 5), lending internal consistency to both determinations. pH rose significantly from 3.92 in PF100 to 5.35 in PF75 ($P < 0.05$), a shift explained by the buffering capacity of insect protein, whose amino, imidazole and guanidino groups consume free hydrogen ions from the lactic, acetic and succinic acids generated during fermentation (Akinyele *et al.*, 2020). Titratable acidity fell correspondingly from 0.29% to 0.22% ($P < 0.05$). The practical implication is mixed: the milder

acidity is likely to be perceived as more palatable by consumers unaccustomed to strongly fermented flavours, but pH values above 4.6 in the three most heavily fortified blends exceed the threshold below which growth of *Clostridium botulinum* and most spoilage bacteria is inhibited, a factor with direct relevance to the microbiological findings in Section 3.8.

Table 5. pH and titratable acidity of pupuru fortified with coconut beetle flour (mean $\pm$ SD, n = 2)

Sample	pH	Titratable acidity (%)
PF100	3.92 ^d $\pm$ 0.02	0.29 ^a $\pm$ 0.00
PF95	4.25 ^d $\pm$ 0.02	0.27 ^a $\pm$ 0.00
PF90	4.59 ^c $\pm$ 0.01	0.25 ^{ab} $\pm$ 0.01
PF85	4.83 ^b $\pm$ 0.01	0.24 ^b $\pm$ 0.00
PF80	5.30 ^a $\pm$ 0.02	0.22 ^b $\pm$ 0.00
PF75	5.35 ^a $\pm$ 0.01	0.22 ^b $\pm$ 0.00

Means within a column bearing different superscript letters differ significantly ($P < 0.05$). Titratable acidity expressed as percentage lactic acid. Sample keys as in Table 1.

3.6 Functional Properties

Bulk density decreased significantly from 0.75 to 0.69 g/mL ($P < 0.05$), attributable to the lower true particle density of insect protein and chitin relative to crystalline cassava starch (Table 6). Water solubility, water absorption capacity and oil absorption capacity all increased significantly with fortification, consistent with the greater hydrophilicity and hydration capacity of insect protein relative to native starch. Least gelation temperature declined from 74.45 °C to 72.65 °C, indicating that the fortified blends gelled at a lower flour concentration, likely due to the additional gelling contribution of insect protein. Swelling power decreased significantly from 9.24% to 7.31% ($P < 0.05$), a consequence of starch dilution compounded by competition between insect protein and starch for available water during heating (Bresciani *et al.*, 2022; Katunzi-Kilewela *et al.*, 2023).

Table 6. Functional properties of pupuru fortified with coconut beetle flour (mean $\pm$ SD, n = 2)

Sample	Bulk density (g/mL)	Water solubility (g/L)	WAC (mL/g)	OAC (mL/g)	LGT (°C)	Swelling power (%)
PF100	0.75 ^a $\pm$ 0.00	6.96 ^c $\pm$ 0.01	1.85 ^b $\pm$ 0.04	1.25 ^b $\pm$ 0.01	74.45 ^a $\pm$ 0.07	9.24 ^a $\pm$ 0.01
PF95	0.74 ^a $\pm$ 0.00	7.20 ^{bc} $\pm$ 0.02	1.94 ^b $\pm$ 0.02	1.32 ^b $\pm$ 0.02	73.80 ^a $\pm$ 0.14	9.08 ^a $\pm$ 0.04
PF90	0.72 ^{ab} $\pm$ 0.00	7.33 ^b $\pm$ 0.02	2.13 ^{ab} $\pm$ 0.01	1.37 ^b $\pm$ 0.01	73.60 ^a $\pm$ 0.14	8.78 ^{ab} $\pm$ 0.01
PF85	0.71 ^b $\pm$ 0.00	7.59 ^{ab} $\pm$ 0.01	2.25 ^a $\pm$ 0.01	1.48 ^{ab} $\pm$ 0.02	73.15 ^{ab} $\pm$ 0.07	8.25 ^b $\pm$ 0.01
PF80	0.70 ^{abc} $\pm$ 0.00	7.74 ^a $\pm$ 0.01	2.34 ^a $\pm$ 0.01	1.54 ^a $\pm$ 0.01	73.00 ^b $\pm$ 0.14	7.95 ^b $\pm$ 0.01
PF75	0.69 ^c $\pm$ 0.01	7.86 ^a $\pm$ 0.01	2.43 ^a $\pm$ 0.03	1.62 ^a $\pm$ 0.01	72.65 ^b $\pm$ 0.07	7.31 ^c $\pm$ 0.02

Means within a column bearing different superscript letters differ significantly ($P < 0.05$). WAC = water absorption capacity; OAC = oil absorption capacity; LGT = least gelation temperature. Sample keys as in Table 1.

3.7 Pasting Properties

With the exception of peak time and pasting temperature, all viscosity parameters decreased significantly with fortification (Table 7), reflecting progressive dilution and physical modification of the starch network. Peak viscosity fell by 57.4%, from 4,274.00 to 1,822.00, while breakdown viscosity fell by 69.3%, the largest proportional change of any pasting parameter, indicating that the fortified pastes were substantially more resistant to thermal and mechanical disintegration than the control. The trough-to-peak stability ratio rose from 0.46 in PF100 to 0.61 in PF75, confirming improved paste stability. Setback viscosity, the standard index of starch retrogradation, fell by 46.3%, implying that fortified pupuru will retain an acceptable texture for longer after cooking than the unfortified product. Peak time and pasting temperature both increased marginally but significantly, from 4.70 to 4.87 minutes and from 74.23 °C to 77.10 °C respectively, consistent with insect protein competing with starch for available water and delaying the onset of gelatinisation (Yang *et al.*, 2025).

Table 7. Pasting properties of pupuru fortified with coconut beetle flour (mean $\pm$ SD, $n = 2$)

Sample	Peak visc.	Trough	Breakdown	Final visc.	Setback	Peak time (min)	Pasting temp (°C)
PF100	4274.00 ^a ± 24.04	1975.50 ^a ± 45.96	2298.50 ^a ± 70.00	2613.00 ^a ± 82.02	667.50 ^a ± 36.06	4.70 ^b ± 0.05	74.23 ^d ± 0.04
PF95	3549.00 ^b ± 24.04	1747.00 ^b ± 50.91	1802.00 ^b ± 74.95	2362.50 ^b ± 61.51	615.50 ^{ab} ± 10.60	4.70 ^b ± 0.05	75.03 ^{cd} ± 0.04
PF90	3138.00 ^c ± 80.61	1726.00 ^b ± 42.43	1412.00 ^c ± 38.18	2272.50 ^b ± 34.65	546.50 ^{abc} ± 7.78	4.73 ^{ab} ± 0.00	75.45 ^{bc} ± 0.57
PF85	2635.50 ^d ± 27.58	1542.50 ^c ± 26.16	1093.00 ^d ± 53.74	2060.50 ^c ± 67.18	518.00 ^{bc} ± 41.01	4.87 ^a ± 0.09	75.80 ^{bc} ± 0.00
PF80	2113.00 ^e ± 32.53	1270.50 ^d ± 30.41	842.50 ^e ± 62.93	1717.50 ^d ± 33.23	447.00 ^{cd} ± 63.64	4.80 ^{ab} ± 0.09	76.20 ^{ab} ± 0.57
PF75	1822.00 ^f ± 46.67	1117.50 ^e ± 53.03	704.50 ^f ± 6.36	1476.00 ^e ± 106.07	358.50 ^e ± 53.03	4.87 ^a ± 0.00	77.10 ^a ± 0.64

Means within a column bearing different superscript letters differ significantly ($P < 0.05$).

Sample keys as in Table 1.

3.8 Microbiological Properties

Fortification produced a mixed microbiological outcome, the only parameter group in which the effect was partly unfavourable (Table 8). Total bacterial count rose significantly, from 9.00×10^2 to 36.00×10^2 cfu/g ($P < 0.05$), driven by the richer nutrient substrate introduced by the insect fraction and by the loss of the acid hurdle documented in Section 3.5, since three of the fortified blends exceeded pH 4.6. Nonetheless, all blends remained well within both the ICMSF (1996) limit of 1×10^6 cfu/g and the stricter Codex Standard 176-1989 limit of 1×10^5 cfu/g for edible cassava flour. Total fungal count, by contrast, fell by 77%, from 6.50×10^2 to 1.50×10^2 cfu/g ($P < 0.05$), plausibly reflecting the antifungal activity of chitin and chitosan in the insect fraction (Muthee *et al.*, 2024; Stull & Weir, 2023), a favourable finding given that moulds are the dominant spoilage and mycotoxin-producing agents in stored cassava flour. all values nonetheless remained at least an order of magnitude below the ICMSF limit of 1×10^4 cfu/g.

Table 8. Microbiological properties of pupuru fortified with coconut beetle flour (mean $\pm$ SD, $n = 2$)

Sample	Total bacterial count ($\times 10^2$ cfu/g)	Total fungal count ($\times 10^2$ cfu/g)
PF100	9.00 ^d $\pm$ 1.41	6.50 ^a $\pm$ 2.12
PF95	17.00 ^c $\pm$ 4.24	6.00 ^b $\pm$ 0.00
PF90	25.00 ^b $\pm$ 2.83	7.00 ^a $\pm$ 2.83
PF85	26.50 ^b $\pm$ 10.61	4.00 ^c $\pm$ 0.00
PF80	33.50 ^a $\pm$ 16.26	2.50 ^d $\pm$ 0.71
PF75	36.00 ^a $\pm$ 1.41	1.50 ^e $\pm$ 2.12

Means within a column bearing different superscript letters differ significantly ($P < 0.05$). Sample keys as in Table 1.

3.9 Sensory Properties

Mean scores for all five sensory attributes declined significantly with increasing substitution (Table 9), illustrating the characteristic tension between nutritional enhancement and consumer acceptability that constrains the fortification of culturally established staple foods. Overall acceptability fell from 8.72 in PF100 to 4.82 in PF75 ($P < 0.05$), with the control and 5% blend rated between like very much and like moderately, the 10% blend corresponding to like slightly, PF85 and PF80 falling into the neither-like-nor-dislike region, and PF75 descending into dislike slightly. Taste and texture declines were mechanistically linked to the amino acid and pasting data: the rise in glutamic and aspartic acid (Section 3.3) introduced a savoury note atypical of a food whose culinary role is to serve as a neutral carrier for soup, while the fall in swelling power and viscosity (Sections 3.6 and 3.7) manifested as a coarser, less cohesive texture. Notably, the 10% blend retained scores between 6.33 and 6.69 across all attributes, closely matching the five-to-seven acceptable range independently reported by Ssepuuya *et al.* (2025) for cricket-fortified cassava flour, corroborating evidence that approximately 10% represents a practical ceiling for insect-based fortification of a cassava swallow food.

Table 9. Sensory properties of pupuru fortified with coconut beetle flour (mean $\pm$ SD, $n = 20$)

Sample	Appearance	Aroma	Taste	Texture	Overall acceptability
PF100	8.61 ^a $\pm$ 1.22	8.57 ^a $\pm$ 1.31	8.53 ^a $\pm$ 1.08	8.67 ^a $\pm$ 1.20	8.72 ^a $\pm$ 1.20
PF95	7.84 ^b $\pm$ 1.22	7.75 ^b $\pm$ 1.31	7.73 ^b $\pm$ 1.08	7.82 ^b $\pm$ 1.20	7.90 ^b $\pm$ 1.20
PF90	6.69 ^c $\pm$ 1.73	6.33 ^c $\pm$ 1.86	6.51 ^c $\pm$ 1.64	6.63 ^c $\pm$ 1.82	6.69 ^c $\pm$ 1.87
PF85	5.86 ^d $\pm$ 1.92	5.49 ^d $\pm$ 1.95	5.63 ^d $\pm$ 1.81	6.12 ^d $\pm$ 1.60	5.98 ^d $\pm$ 1.61
PF80	5.88 ^d $\pm$ 1.95	5.33 ^d $\pm$ 2.09	5.31 ^{de} $\pm$ 2.09	6.10 ^d $\pm$ 1.80	5.67 ^d $\pm$ 1.98
PF75	5.20 ^e $\pm$ 2.24	4.35 ^e $\pm$ 2.73	4.49 ^e $\pm$ 2.82	4.82 ^e $\pm$ 2.46	4.82 ^e $\pm$ 2.50

Means within a column bearing different superscript letters differ significantly ($P < 0.05$). Scores based on a nine-point hedonic scale (1 = dislike extremely, 9 = like extremely). Sample keys as in Table 1.

4. Conclusion

Fortification of pupuru with coconut beetle (*Oryctes rhinoceros*) larvae flour produced a substantial and largely favourable transformation of its nutritional, functional and safety profile, raising crude protein more than fivefold, enriching calcium, potassium, zinc and iron, improving the amino acid score from approximately 49% to 73%, and reducing tannin and cyanogenic glycoside, while total fungal counts fell and pasting stability improved. These gains were accompanied by a progressive decline in sensory acceptability and a rise in total bacterial counts at higher substitution levels, underscoring the need for strict hygienic control during insect flour preparation. Integrating the nutritional, safety and sensory findings, the 10% substitution blend (PF90) emerged as the optimal formulation, more than doubling protein content while retaining an overall acceptability score corresponding to like slightly. This blend represents a practical, culturally compatible route to improving the nutritional quality of a widely consumed cassava staple, though further work on storage stability, protein and mineral bioavailability, and consumer acceptance within pupuru-consuming communities is needed before the formulation can be recommended for wider adoption.

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